

Chapter 5: Equations

Three-week revision booklet • 6 worksheets • 300 questions

NSW syllabus outcome

MA4-EQU-C-01 — solves linear equations of up to two steps •
MAO-WM-01

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How to use this booklet

There are six worksheets, two for each week. Each one drills a single skill from the Equations topic test and now runs across **two pages** with **50 questions** and room to show your working. Each sheet is split into **Easy**, **Medium** and **Challenging**. Every section starts with a worked example that solves the equation two ways — by **balancing the sides** and by using **inverse operations** — so you can use whichever method makes sense to you. One-step questions have one working line, two-step questions have two. Work from the top of the page down; the questions get harder as you go. The final worksheet mixes every skill together, so it is the best check that you are ready for the topic test.

Week	Worksheet	Skill
1	1	One-step equations with + and –
1	2	One-step equations with $\times$ and $\div$
2	3	Two-step equations
2	4	Equations with fractions
3	5	Writing and solving equations
3	6	Mixed revision — all skills

Progress tracker

Worksheet	Date completed	Score / 50	Checked by
1 One-step equations with + and –			
2 One-step equations with $\times$ and $\div$			
3 Two-step equations			
4 Equations with fractions			
5 Writing and solving equations			
6 Mixed revision — all skills			

Worksheet 1: One-step equations: + and -

Week 1 · Worksheet 1 of 6 · 50 questions

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Do the same thing to both sides. To undo an addition, subtract; to undo a subtraction, add.

EASY

30 QUESTIONS

WORKED EXAMPLE

$$x + 5 = 12$$

BALANCING THE SIDES

$$x + 5 = 12$$

$$x + 5 - 5 = 12 - 5$$

$$x = 7$$

INVERSE OPERATIONS

The inverse of "+ 5" is "- 5".

$$x = 12 - 5$$

$$x = 7$$

Solve each equation.

1. $w + 17 = 52$

2. $e - 12 = 0$

3. $x + 5 = 14$

4. $y - 8 = 3$

5. $a + 11 = 20$

6. $b - 6 = 9$

7. $m + 4 = 4$

8. $n - 15 = 5$

9. $p + 23 = 40$

10. $q - 7 = 12$

11. $c + 9 = 25$

12. $d - 3 = 18$

13. $g + 30 = 45$

14. $h - 21 = 9$

15. $j + 6 = 31$

16. $k - 14 = 6$

17. $r + 8 = 8$

18. $s - 25 = 25$

19. $t + 19 = 33$

20. $u - 11 = 29$

21. $f + 13 = 29$

22. $v - 9 = 16$

23. $z + 21 = 21$

24. $i - 4 = 11$

25. $o + 16 = 24$

26. $x + 6 = 19$

27. $y - 5 = 12$

28. $m + 14 = 30$

29. $n - 18 = 2$

30. $p + 7 = 40$

Worksheet 1: One-step equations: + and -

Week 1 · Worksheet 1 of 6 · continued

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MEDIUM

12 QUESTIONS

WORKED EXAMPLE

$$x + 9 = 4$$

BALANCING THE SIDES

$$x + 9 = 4$$

$$x + 9 - 9 = 4 - 9$$

$$x = -5$$

INVERSE OPERATIONS

The inverse of "+ 9" is "- 9".

$$x = 4 - 9$$

$$x = -5$$

31. $m + 9 = 4$

32. $n - 3 = -10$

33. $p + 14 = 6$

34. $q - 8 = -2$

35. $x + 7 = 0$

36. $y - 5 = -12$

37. $a + 20 = 11$

38. $b - 1 = -9$

39. $c + 15 = 15$

40. $d - 6 = -6$

41. $e + 4 = -3$

42. $g - 12 = -20$

CHALLENGING

8 QUESTIONS

WORKED EXAMPLE

$$n - 6 = -2$$

BALANCING THE SIDES

$$n - 6 = -2$$

$$n - 6 + 6 = -2 + 6$$

$$n = 4$$

INVERSE OPERATIONS

The inverse of "- 6" is "+ 6".

$$n = -2 + 6$$

$$n = 4$$

43. Solve $x + 3.5 = 9$

44. Solve $y - \frac{1}{2} = \frac{3}{4}$

45. I think of a number and add 17. The answer is 43. Find the number.

46. I think of a number and subtract 9. The answer is -4. Find the number.

47. The perimeter of a triangle is 24 cm. Two sides are 9 cm and 7 cm. Find the third side.

48. A bank balance rose by \$85 to \$210. Write and solve an equation for the original balance.

49. Solve $12 = n - 8$

50. Check whether $x = 6$ is a solution of $x + 9 = 16$. Show your working.

Worksheet 2: One-step equations: $\times$ and $\div$

Week 1 · Worksheet 2 of 6 · 50 questions

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To undo a multiplication, divide both sides. To undo a division, multiply both sides.

EASY

30 QUESTIONS

WORKED EXAMPLE

$4x = 20$

BALANCING THE SIDES

$$4x = 20$$

$$4x \div 4 = 20 \div 4$$

$$x = 5$$

INVERSE OPERATIONS

The inverse of " $\times 4$ " is " $\div 4$ ".

$$x = 20 \div 4$$

$$x = 5$$

Solve each equation.

1. $5p = 15$

2. $3x = 12$

3. $7y = 42$

4. $4m = 36$

5. $9n = 63$

6. $6a = 48$

7. $8b = 24$

8. $2c = 26$

9. $10d = 90$

10. $12e = 60$

11. $\frac{x}{2} = 6$

12. $\frac{m}{4} = 7$

13. $\frac{y}{3} = 5$

14. $\frac{n}{10} = 4$

15. $\frac{a}{6} = 2$

16. $\frac{b}{5} = 9$

17. $\frac{c}{8} = 3$

18. $\frac{d}{7} = 1$

19. $\frac{e}{9} = 6$

20. $\frac{g}{12} = 2$

21. $6p = 54$

22. $11m = 44$

23. $\frac{y}{6} = 8$

24. $4x = 100$

25. $13c = 39$

26. $4k = 28$

27. $\frac{p}{3} = 7$

28. $9x = 81$

29. $\frac{q}{4} = 5$

30. $7m = 56$

Worksheet 2: One-step equations: $\times$ and $\div$

Week 1 · Worksheet 2 of 6 · continued

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MEDIUM

12 QUESTIONS

WORKED EXAMPLE

$$-3x = 18$$

BALANCING THE SIDES

$$-3x = 18$$

$$-3x \div (-3) = 18 \div (-3)$$

$$x = -6$$

INVERSE OPERATIONS

The inverse of " $\times (-3)$ " is " $\div (-3)$ ".

$$x = 18 \div (-3)$$

$$x = -6$$

31. $3k = -12$

32. $-2m = -16$

33. $-5n = 40$

34. $6p = -54$

35. $\frac{x}{3} = -4$

36. $\frac{y}{-2} = 7$

37. $-4a = 28$

38. $8b = -72$

39. $\frac{m}{5} = -3$

40. $-9c = -45$

41. $7d = -21$

42. $\frac{n}{-6} = -2$

CHALLENGING

8 QUESTIONS

WORKED EXAMPLE

$$\frac{x}{5} = 3$$

BALANCING THE SIDES

$$\frac{x}{5} = 3$$

$$\frac{x}{5} \times 5 = 3 \times 5$$

$$x = 15$$

INVERSE OPERATIONS

The inverse of " $\div 5$ " is " $\times 5$ ".

$$x = 3 \times 5$$

$$x = 15$$

43. Solve $2.5x = 10$

44. Solve $\frac{2}{3}y = 8$

45. Six equal parcels weigh 27 kg in total. Find the mass of one parcel.

46. A number divided by 7 gives 12. Find the number.

47. A rectangle has area 84 cm^2 and width 6 cm. Write and solve an equation for its length.

48. Tickets cost \$9 each. Total takings were \$342. Write and solve an equation for the number sold.

49. Solve $-36 = 4x$

50. Check whether $m = -3$ is a solution of $-5m = 15$. Show your working.

Worksheet 3: Two-step equations

Week 2 · Worksheet 3 of 6 · 50 questions

NSW syllabus outcome

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Undo the addition or subtraction first, then undo the multiplication or division.

EASY

30 QUESTIONS

WORKED EXAMPLE

$$2x + 3 = 11$$

BALANCING THE SIDES

$$2x + 3 = 11$$

$$2x + 3 - 3 = 11 - 3$$

$$2x = 8$$

$$2x \div 2 = 8 \div 2$$

$$x = 4$$

INVERSE OPERATIONS

Undo "+ 3" first: subtract 3.

$$2x = 8$$

Undo "× 2" next: divide by 2.

$$x = 4$$

Solve each equation.

1. $3t + 4 = 10$

2. $2n + 6 = 12$

3. $4y - 7 = 5$

4. $5x + 3 = 18$

5. $6m - 2 = 22$

6. $2a + 9 = 15$

7. $7b - 5 = 16$

8. $3c + 8 = 20$

9. $4d - 6 = 14$

10. $8e + 1 = 17$

11. $5g - 12 = 8$

12. $9h + 4 = 40$

13. $2j - 11 = 3$

14. $6k + 7 = 25$

15. $3p - 9 = 0$

16. $10q + 5 = 55$

17. $4r - 3 = 21$

18. $7s + 6 = 34$

19. $5t - 15 = 10$

20. $2u + 13 = 21$

21. $6v + 9 = 33$

22. $3w - 4 = 11$

23. $8x + 2 = 26$

24. $5y - 6 = 19$

25. $4z + 10 = 30$

26. $2a + 7 = 19$

27. $5b - 4 = 26$

28. $3c + 11 = 29$

29. $7d - 2 = 19$

30. $4e + 6 = 22$

Worksheet 3: Two-step equations

Week 2 • Worksheet 3 of 6 • continued

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MEDIUM

12 QUESTIONS

WORKED EXAMPLE

$$3x + 7 = -2$$

BALANCING THE SIDES

$$3x + 7 = -2$$

$$3x + 7 - 7 = -2 - 7$$

$$3x = -9$$

$$x = -3$$

INVERSE OPERATIONS

Undo "+ 7": subtract 7.

$$3x = -9$$

Undo "× 3": divide by 3.

$$x = -3$$

31. $2r - 5 = 9$

32. $3x + 11 = 2$

33. $4y - 1 = -13$

34. $5m + 8 = -7$

35. $6n - 4 = -28$

36. $2p + 15 = 5$

37. $7q - 9 = -30$

38. $3a + 20 = 2$

39. $8b - 6 = -38$

40. $4c + 17 = 1$

41. $9d - 2 = -29$

42. $5e + 12 = -8$

CHALLENGING

8 QUESTIONS

WORKED EXAMPLE

$$2(x + 3) = 14$$

BALANCING THE SIDES

$$2(x + 3) = 14$$

$$2(x + 3) \div 2 = 14 \div 2$$

$$x + 3 = 7$$

$$x = 4$$

INVERSE OPERATIONS

Undo "× 2" first: divide by 2.

$$x + 3 = 7$$

Undo "+ 3": subtract 3.

$$x = 4$$

43. $10 - 2x = 4$

44. $15 - 3y = 6$

45. $7 - 4m = -9$

46. $20 - 5n = 0$

47. $2(x + 3) = 14$

48. $3(y - 2) = 9$

49. I think of a number, multiply by 4 and add 7. The answer is 31. Find the number.

50. A taxi charges \$5 plus \$3 per km. A trip cost \$29. Write and solve an equation for the distance.

Worksheet 4: Equations with fractions

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Multiply both sides by the denominator to clear the fraction, then solve as usual.

EASY

30 QUESTIONS

WORKED EXAMPLE

$$\frac{x}{3} + 2 = 5$$

BALANCING THE SIDES

$$\frac{x}{3} + 2 = 5$$

$$\frac{x}{3} + 2 - 2 = 5 - 2$$

$$\frac{x}{3} = 3$$

$$\frac{x}{3} \times 3 = 3 \times 3$$

$$x = 9$$

INVERSE OPERATIONS

Undo “+ 2” first: subtract 2.

$$\frac{x}{3} = 3$$

Undo “÷ 3” next: multiply by 3.

$$x = 9$$

Solve each equation.

1. $\frac{p}{3} + 2 = 7$

5. $\frac{n}{6} - 2 = 4$

9. $\frac{d}{7} - 3 = 0$

13. $\frac{y-1}{5} = 2$

17. $\frac{b-3}{6} = 2$

21. $\frac{f}{4} + 3 = 6$

25. $\frac{m}{5} + 8 = 9$

29. $\frac{q-6}{2} = 5$

2. $\frac{x}{2} + 5 = 9$

6. $\frac{a}{3} + 7 = 10$

10. $\frac{e}{8} + 1 = 5$

14. $\frac{m+6}{4} = 3$

18. $\frac{c+9}{5} = 4$

22. $\frac{g}{9} - 1 = 2$

26. $\frac{x}{3} + 4 = 9$

30. $\frac{r}{8} + 2 = 4$

3. $\frac{m}{4} - 1 = 3$

7. $\frac{b}{2} - 6 = 1$

11. $\frac{q-4}{2} = 3$

15. $\frac{n-5}{2} = 6$

19. $\frac{d-8}{3} = 1$

23. $\frac{h+3}{4} = 5$

27. $\frac{n}{5} - 2 = 3$

4. $\frac{y}{5} + 3 = 8$

8. $\frac{c}{10} + 4 = 6$

12. $\frac{x+2}{3} = 4$

16. $\frac{a+1}{7} = 1$

20. $\frac{e+4}{2} = 9$

24. $\frac{k-2}{6} = 3$

28. $\frac{p+2}{4} = 3$

Worksheet 4: Equations with fractions

Week 2 · Worksheet 4 of 6 · continued

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MEDIUM

12 QUESTIONS

WORKED EXAMPLE

$$\frac{x-1}{4} = 3$$

BALANCING THE SIDES

$$\frac{x-1}{4} = 3$$

$$\frac{x-1}{4} \times 4 = 3 \times 4$$

$$x-1 = 12$$

$$x = 13$$

INVERSE OPERATIONS

Undo "÷ 4": multiply by 4.

$$x-1 = 12$$

Undo "- 1": add 1.

$$x = 13$$

31. $\frac{2x}{5} = 4$

34. $\frac{4n}{3} = 8$

37. $\frac{3b-2}{4} = 4$

40. $\frac{y-3}{-2} = 4$

32. $\frac{3y}{4} = 9$

35. $\frac{x}{3} + 2 = -1$

38. $\frac{m}{2} - 7 = -10$

41. $\frac{7d}{2} = 21$

33. $\frac{5m}{2} = -15$

36. $\frac{2a+1}{3} = 5$

39. $\frac{5c}{6} = 10$

42. $\frac{n+5}{-3} = 2$

CHALLENGING

8 QUESTIONS

WORKED EXAMPLE

$$\frac{x}{2} + \frac{x}{3} = 5$$

BALANCING THE SIDES

$$\frac{x}{2} + \frac{x}{3} = 5$$

× 6 (the LCD) both sides:

$$3x + 2x = 30$$

$$5x = 30$$

$$x = 6$$

INVERSE OPERATIONS

Fractions with different denominators can't be undone one at a time - clear them first by multiplying by the LCD, then solve normally.

43. Solve $\frac{x}{2} + \frac{x}{3} = 5$

45. One-third of a number, plus 6, is 14. Find the number.

47. A recipe uses $\frac{1}{4}$ of a bag of flour per cake. Eight cakes use 2 bags. Check this with an equation.

49. The average of a number and 12 is 19. Find the number.

44. Solve $\frac{2m-1}{3} = \frac{m+4}{2}$

46. Half a number take away 5 gives 12. Find the number.

48. Solve $\frac{x+1}{2} = \frac{x-3}{4}$

50. Explain why multiplying both sides of $\frac{x}{5} = 3$ by 5 is a valid step.

Worksheet 5: Writing and solving equations

Week 3 · Worksheet 5 of 6 · 50 questions

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Choose a pronumeral for the unknown, write an equation, then solve it and answer in words.

EASY

30 QUESTIONS

WORKED EXAMPLE

"A number plus 4 is 9."

Let the number be n .

"plus 4" means $+ 4$ "is 9" means $= 9$

Equation: $n + 4 = 9$

Write an equation for each statement. Use n .

1. A number times 12 is 156.

2. A number plus 8 is 21.

3. A number minus 5 is 17.

4. Three times a number is 24.

5. A number divided by 4 is 9.

6. Twice a number, plus 3, is 19.

7. Five less than a number is 11.

8. A number doubled is 30.

9. Four times a number, less 6, is 18.

10. Half a number is 13.

11. A number increased by 15 is 40.

12. Seven times a number is 63.

13. A number decreased by 9 is 2.

14. One-third of a number is 6.

15. Three more than twice a number is 17.

16. A number added to itself is 26.

17. Six times a number, plus 1, is 37.

18. A number less 12 is 0.

19. Ten times a number is 250.

20. Nine less than three times a number is 12.

21. A number multiplied by 5 is 45.

22. Eleven more than a number is 20.

23. A number divided by 6 is 3.

24. Triple a number, minus 4, is 11.

25. A number take away 7 is 15.

26. A number plus 6 is 14.

27. Double a number, less 5, is 9.

28. A number divided by 3 is 7.

29. Eight times a number is 72.

30. Four less than a number is 10.

Worksheet 5: Writing and solving equations

Week 3 • Worksheet 5 of 6 • continued

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MEDIUM

12 QUESTIONS

WORKED EXAMPLE

Ella is 5 years older than Ben. Together they are 21.

BALANCING THE SIDES

Let Ben's age be b , so Ella is $b + 5$.
 $b + (b + 5) = 21$
 $2b + 5 = 21$, so $2b = 16$, so $b = 8$

INVERSE OPERATIONS

Undo "+ 5": subtract 5 from both sides.
 $2b = 16$
 Undo " $\times 2$ ": divide both sides by 2.
 $b = 8$

1. Ella is 3 years older than Sam. Together they are 27. How old is Sam?

4. Three equal boxes plus 2 kg balance 20 kg. Find one box.

7. A number less 8, then halved, gives 6. Find the number.

10. A plan costs \$20 plus \$0.50 per GB. The bill is \$35. Find the GB used.

2. Four consecutive numbers add to 34. Find the smallest.

5. The sum of a number and its double is 42. Find the number.

8. Two numbers differ by 7 and add to 33. Find both.

11. Three shirts and a \$15 hat cost \$90. Find the shirt price.

3. A number times 7, then less 4, gives 45. Find the number.

6. A class has 5 more girls than boys and 29 students. Find the boys.

9. A rectangle is twice as long as it is wide, perimeter 36 cm. Find the width.

12. A taxi charges \$4 plus \$2.50 per km. A trip cost \$19. Find the distance.

CHALLENGING

8 QUESTIONS

WORKED EXAMPLE

A taxi charges \$6 plus \$2 per km. A trip cost \$18.

BALANCING THE SIDES

Let the distance be d km.
 $6 + 2d = 18$
 $2d = 12$, so $d = 6$

INVERSE OPERATIONS

Undo "+ 6": subtract 6 from both sides.
 $2d = 12$
 Undo " $\times 2$ ": divide both sides by 2.
 $d = 6$

13. A raffle costs \$150 for prizes. Tickets sell for \$6. How many just cover costs?

15. A rectangle is 4 cm longer than it is wide, perimeter 48 cm. Find both sides.

17. A plumber charges \$80 plus \$65 per hour. A job cost \$340. Find the hours.

19. A number times 12 gives 156. Which fits: $n + 12 = 156$, $12n = 156$ or $156 - n = 12$? Solve it.

14. To make \$300 profit as well, how many \$6 tickets are needed in total?

16. Five consecutive numbers add to 60. Find them.

18. The angles of a triangle are x° , $2x^\circ$ and $3x^\circ$. Find all three.

20. A number is 3 more than another. Their sum is 41. Find both.

Worksheet 6: Mixed revision

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This sheet mixes every skill from the Equations chapter. Work through it as though it were the topic test.

● EASY

30 QUESTIONS

WORKED EXAMPLE

$$3x - 4 = 11$$

BALANCING THE SIDES

$$3x - 4 = 11$$

$$3x - 4 + 4 = 11 + 4$$

$$3x = 15$$

$$x = 5$$

INVERSE OPERATIONS

Undo “− 4”: add 4.

$$3x = 15$$

Undo “× 3”: divide by 3.

$$x = 5$$

Solve each equation.

1. $w + 17 = 52$

5. $3t + 4 = 10$

9. $m + 9 = 4$

13. $\frac{m}{5} = 4$

17. $9 + x = 2$

21. $n - 22 = 5$

25. $\frac{c}{2} - 3 = 4$

29. $4n - 5 = 19$

2. $e - 12 = 0$

6. $2n + 6 = 12$

10. $3k = -12$

14. $2r - 5 = 9$

18. $\frac{y}{4} - 1 = 2$

22. $9m = 108$

26. $x + 12 = 30$

30. $\frac{p}{2} + 6 = 10$

3. $5p = 15$

7. $4y - 7 = 5$

11. $8a = 56$

15. $6c + 3 = 21$

19. $7n = -35$

23. $\frac{a}{7} = 8$

27. $6y = 48$

27. $6y = 48$

4. $\frac{x}{2} = 6$

8. $\frac{p}{3} + 2 = 7$

12. $b - 14 = 6$

16. $-4d = 24$

20. $5g - 8 = 12$

24. $3x + 5 = 26$

28. $\frac{m}{3} = 9$

28. $\frac{m}{3} = 9$

Worksheet 6: Mixed revision

Week 3 • Worksheet 6 of 6 • continued

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● MEDIUM

12 QUESTIONS

WORKED EXAMPLE

$$2(x - 1) = 8$$

BALANCING THE SIDES

$$2(x - 1) = 8$$

$$2(x - 1) \div 2 = 8 \div 2$$

$$x - 1 = 4$$

$$x = 5$$

INVERSE OPERATIONS

Undo "x 2": divide by 2.

$$x - 1 = 4$$

Undo "- 1": add 1.

$$x = 5$$

31. $\frac{a-4}{2} = 3$

34. $\frac{2m}{5} = 6$

37. $\frac{n+5}{3} = 4$

40. $\frac{3d}{4} = 9$

32. $10 - 2x = 4$

35. $2(x + 3) = 14$

38. $15 - 3b = 6$

41. $5e - 2 = -17$

33. $3y + 11 = 2$

36. $4a - 9 = -21$

39. $6c + 7 = 1$

42. $8 - 4g = 0$

● CHALLENGING

8 QUESTIONS

WORKED EXAMPLE

A number is halved, then 3 is added, giving 9.

BALANCING THE SIDES

Let the number be n .

$$\frac{n}{2} + 3 = 9$$

$$\frac{n}{2} = 6$$

$$n = 12$$

INVERSE OPERATIONS

Undo "+ 3": subtract 3.

$$\frac{n}{2} = 6$$

Undo " $\div 2$ ": multiply by 2.

$$n = 12$$

43. A charity raffle costs \$150 for prizes. Tickets sell for \$6. How many tickets just cover costs?

45. A number is multiplied by 12 giving 156. Write the equation and find the number.

47. A rectangle's perimeter is 40 cm and its length is 12 cm. Find its width.

49. A taxi charges \$5 plus \$3 per km. A trip cost \$29. Find the distance.

44. To make \$300 profit on top of that \$150, how many \$6 tickets must be sold in total?

46. I think of a number, multiply by 5 and subtract 8. The answer is 27. Find the number.

48. Three consecutive numbers add to 48. Find them.

50. Solve the guess, check and improve way: $4m - 9 = 7$. Show at least two attempts.
